Sanjukktha Senthil Kumar

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EDUCATION

Master of Science in Computer Science, University of Texas at Arlington | Arlington, TX **Aug 2024 - May 2026**
Software Engineering II | Data Analysis & Modelling Techniques | Design and Analysis of Algorithm ***(GPA: 4.00/4.00)***

Bachelor of Technology in Computer Science and Engineering, SASTRA University | Tamil Nadu, India
Software Engineering | Data warehouse & Data mining | Natural Language Processing | Cloud Computing ***(GPA: 4.00/4.00)***

WORK EXPERIENCE

Goldman Sachs – Summer Engineer Analyst | Dallas, Texas, USA **Jun 2025 - Aug 2025**

- Architected TalentFlowAI, a multi-step Agentic AI pipeline using LangChain AgentExecutor with Pydantic-validated tools, GPT-4 via Azure OpenAI and cosine similarity search over Pinecone vector embeddings to automate candidate screening
- Engineered a Python/FastAPI microservice for AI inference integrated with Spring Boot backend over HTTP, following enterprise polyglot architecture patterns for production AI/ML in Java stacks
- Implemented production reliability guardrails including iteration caps, automatic retry on malformed tool calls, Pydantic input schema validation, and explicit zero-result handling to prevent hallucinated candidates
- Built a candidate progression tracking platform in React/TypeScript with a Spring Boot/MongoDB backend, surfacing real-time pipeline blockers across pre-interview vetting, offer, and post-offer stages in the candidate selection process
- Integrated ServiceNow ticket visibility at pipeline blockers; built reusable WCAG-compliant UI components using GS UI toolkit to ensure compliance with accessibility standards

Saint-Gobain Research – Product Engineer, Research Systems | Tamil Nadu, India **Jan 2024 - Jun 2024**

- Led a Phase 0 feasibility study evaluating AI automation potential for experiment intake, designing a OpenAI extraction layer with Tool calling to parse unstructured submissions into JSON records, quantifying 800+ requests/2 wks with 76% of QA tasks automatable
- Built an analytics platform with Node.js/REST backend and Python ingestion pipelines supporting comparison across 30+ material scenarios and 15 experimental use cases, enabling researchers to cut manual aggregation
- Designed Pydantic schema with bulk insert via SQLAlchemy, enforcing canonical data standards across 8 global sites
- Integrated platform with Salesforce (sObject API) for technical reporting and to surface material data directly on client records

TeleApps – Associate Product Operations Engineer | Tamil Nadu, India **Mar 2022 - Dec 2023**

- Engineered ML inference services using Kafka, FastAPI, Redis caching and ONNX Runtime to classify caller intent in IVR-Utterances for a telecom client, achieving sub-200ms end-to-end latency under concurrent campaign workloads without GPU infrastructure
- Containerized microservices for inference with request-level caching, model versioning, and lazy loading at first inference request
- Designed ETL pipelines in pandas including lag features, rolling windows, seasonality indicators and trend decomposition for a call-volume workload prediction model reducing customer wait times

TeleApps –Technology Operations Analyst | Tamil Nadu, India **Jun 2021 - Feb 2022**

- Built Python data pipelines to extract, validate, and reconcile wind turbine sensor time-series data, implementing per-column z-score outlier detection and forward-fill gap recovery to preserve data integrity without silently dropping flagged readings
- Developed performance dashboards using matplotlib and seaborn including heatmaps and anomaly scatter plots, enabling operations teams to identify underperforming turbines
- Configured Adobe Workfront project workflows with task dependencies, approval stages, and cross-team status tracking to coordinate batched data deliverables to client-facing delivery teams

PROJECTS

High-Frequency Event Queue Simulator **Mar 2025**

- Designed a distributed event simulation platform processing 50K+ events per minute, using Kafka topics with partitioned producers and Redis backed coordination to simulate real world queue bursts, retries, and backpressure under peak load conditions
- Engineered real-time dashboard in React with WebSocket updates to monitor throughput, retries, and failure rates under heavy load
- Benchmarked and optimized end to end streaming pipelines by tuning Kafka batch sizes, consumer concurrency, and Redis access patterns, achieving sub 200ms end to end latency while maintaining delivery guarantees in high volume scenarios

Mini-Okta - Secure Authentication Service **Feb 2024**

- Developed a secure authentication and MFA service using Node.js, JWT, and OAuth2, supporting token refresh workflows
- Implemented role-based access control (RBAC) middleware for granular user and admin-level permissions
- Integrated Redis based caching for active token state and session validation, optimizing token lookup, reducing authentication latency by ~9% under concurrent request loads

TECHNICAL SKILLS

- **Programming Languages**: Python, C, C++, JAVA, R, Ruby, JavaScript, TypeScript, Go, Rust, Bash/Shell Scripting
- **Tools**: Git, RESTful APIs, GraphQL, Docker, AWS, Azure, Jenkins, Kubernetes, MongoDB, MySQL, Spring Boot, Jira
- **Web Dev**: React.js, Node.js, HTML5, Django, Flask, Ruby on Rails, Next.js, Tailwind CSS
- **AI Dev**: MCP, Claude Code, LangGraph, LangChain, RAG, OpenAI, GPT-4, Tool Calling, Whisper, Hugging Face, LlamaIndex, n8n
- **Knowledge areas**: SDLC, Version Control Systems, API Design, Software Architecture, Cross-Platform Development

ACHIEVEMENTS

- **HackUTA 2024** - University of Texas at Arlington: Developed “Attention Seeker”, a Chrome extension to improve task focus for users with attention challenges by combining real-time webcam analysis and screen activity monitoring; implemented OpenCV-based gaze detection, browser APIs for activity tracking, and a Python backend for behavioral scoring, delivering adaptive alerts that improved sustained focus during testing sessions by 34%
- **Certifications**: AWS Certified Cloud Practitioner, AWS Certified Solutions Architect, Stanford Machine Learning Certified, Microsoft Azure Data Engineer Associate, Microsoft SQL Server Certification, SnowPro Core Certification, Azure Fundamentals